Circles Determined by Planar Point Sets

Erdős Problem 506

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Abstract

For $n \geq 4$, let $c(n)$ be the minimum number of distinct circles containing at least three points of an n-point set in the Euclidean plane, where the set is neither collinear nor concyclic. Put

$$F(n) = 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor .$$

We determine $c(n)$ for every $n \geq 4$: it equals $F(n)$ apart from three exceptional orders. We also solve the variant in which no three points are collinear; that variant has a single exceptional order. The proofs and exact finite verifications were developed through a collaboration between human researchers and artificial-intelligence systems.

Keywords: planar point sets; concyclic points; inversion; line arrangements; exact computer verification

1 Introduction

Definition 1.1. Let $P \subset \mathbb{R}^2$ be finite, and let $c(P)$ denote the number of distinct Euclidean circles containing at least three points of P. For $n \geq 4$, define

$$c(n) = \min_{P} c(P),$$

where the minimum is over all n-point sets P that are neither collinear nor concyclic. We call a finite point set satisfying these two conditions *admissible*. A line is not counted as a circle, and a circle containing $k \geq 3$ points is counted once.

Take $n-1$ points on a circle Γ and a point p outside Γ. The points on Γ can be grouped into $\lfloor (n-1)/2 \rfloor$ pairs such that each pair is collinear with p. Apart from Γ, every other pair of points on Γ that is not collinear with p determines a circle with p. Distinct pairs give distinct circles, because two distinct circles have at most two common points. Thus

$$c(n) \leq F(n) := 1 + \binom{n-1}{2} - \left\lfloor \frac{n-1}{2} \right\rfloor . \tag{1}$$

Elliott initiated a systematic study of this problem in 1967 [6]. His Theorem 1 proves that every admissible set of $n > 3$ points determines at least

$$\frac{2n(n-1)}{63}$$

circles containing exactly three points; such circles are usually called *ordinary circles*. His Theorem 2 asserts that the total number of determined circles is at least $\binom{n-1}{2}$ for $n > 393$ [6, Theorems 1–2]. Purdy and Smith identified the missing subtraction caused by the pairs made

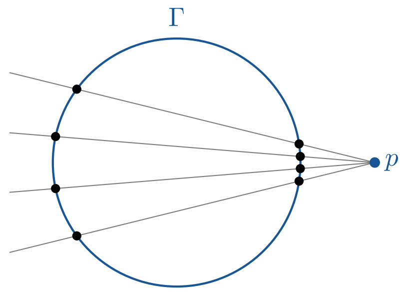

Figure 1: The benchmark construction, illustrated for $n = 9$. Four secants through p pair the eight points of Γ; those four collinear pairs do not produce circles through p.

collinear with the exterior point in figure 1. They observed that Elliott's argument can be modified, without changing the threshold, to give $c(n) \geq F(n)$ for $n \geq 394$; together with (1), this yields $c(n) = F(n)$ in that range [9, §2.1].

Bálintová and Bálint proved the general bound

$$c(P) \geq \frac{15n(n-1)+1678}{266} \qquad (n \geq 6)$$

in their Theorem 2.4, using the pointed ordinary-circle estimate of their Theorem 2.1 and the six-point initial case in Lemma 6.1 [1, Theorems 2.1 and 2.4 and Lemma 6.1]. A related but distinct line of work counts only ordinary circles. Zhang's Theorem 4.1 gives at least $m/6$ ordinary lines avoiding a prescribed external point; inversion and double counting then yield the ordinary-circle bound $\frac{1}{9}\binom{n}{2}$ in his Theorem 4.2 [12, Theorems 4.1–4.2]. Lin et al. later determined the exact minimum number of ordinary circles for sufficiently large n [8, Theorem 1.1(i)]. These results provide useful tools, but they do not directly settle the present extremal problem: a circle containing many points is counted once here, although it contains many triples.

The results certified in this paper are as follows.

Theorem 1.2. *For every $n \geq 4$, one has $c(n) = F(n)$, with precisely three exceptions:*

$$c(6) = 8 = F(6) - 1, \qquad c(7) = 11 = F(7) - 2, \qquad c(8) = 17 = F(8) - 2.$$

Elliott records Segre's observation that a planar projection of a cube gives an eight-point counterexample to Elliott's proposed bound [6, p. 182, discussion following Theorem 2]. That configuration determines 18 circles. We prove that the first exceptional order already occurs at $n = 6$, and we improve the eight-point construction to one with $c(P) = 17$. Explicit constructions for the exceptional cases are shown in figures 2 to 4.

Let $c_{\mathrm{nc}}(n)$ denote the minimum of $c(P)$ over admissible n-point sets having no three collinear points. The additional hypothesis changes the answer only once.

Corollary 1.3. *For every $n \geq 4$,*

$$c_{\mathrm{nc}}(n) = 1 + \binom{n-1}{2},$$

except that $c_{\mathrm{nc}}(8) = 20$.

Purdy and Smith's correction of Elliott's argument settles every $n \geq 394$ and therefore makes the unresolved part finite in principle [9, §2.1]. In practice, however, the remaining incidence structures form a very large and irregular search space, for which a direct verification is neither small nor naturally uniform. The present proof is independent of that large-n threshold: it establishes the result for every $n \geq 15$, rather than merely filling the interval $15 \leq n \leq 393$. We also solve the no-three-collinear variant. For the original problem with $n \leq 14$, exact computer verification is applied only after mathematical constraints have reduced the possible incidence data.

The project was developed in several stages through collaboration between the author and artificial-intelligence systems. The author first used Eureka, an AI-assisted mathematical research system developed by the JiuChong team at the University of Science and Technology of China, to obtain numerical results for every $n \geq 16$. Examination of those computations showed that the range $n \geq 92$ was already controlled by inequalities that could be distilled almost entirely into a mathematical argument; that earlier argument is included in Appendix A. Starting from it, the author worked with OpenAI Codex to strengthen the estimates until they yielded the proof for every $n \geq 15$. Codex was also used to complete the exact verification of all remaining small orders and to search for exceptional configurations. The mathematical part of the argument has been translated into Lean 4 and checked against Mathlib. The formalization and exact-verification files are available in the accompanying GitHub repository `https://github.com/LyonWang00/Erdos-problem/tree/main/Erdos506`.

2 Notation and basic tools

Fix an admissible n-point set P. Let K be the largest number of points of P on a line or a circle, and put $r = n - K$. A line or circle attaining K will be called a *largest line* or *largest circle*, respectively. Every maximal collinear or concyclic subset of P containing at least three points will be called, respectively, a *line block* or a *circle block*; when the distinction is immaterial, either is called a *block*.

Whenever the points are labelled $p_0, p_1, \ldots$, a string of indices denotes the corresponding subset; for example, 0256 means $\{p_0, p_2, p_5, p_6\}$. In an incidence list the surrounding text specifies whether the subset is collinear or concyclic. This convention is used only to shorten finite lists; it does not identify a circle with an index string unless at least three of the indicated points are noncollinear.

For the classification of the exceptional configurations we need two precise notions of incidence type. A *generalized circle* is a Euclidean line or a Euclidean circle, equivalently a circle on the Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. Let $\mathcal{B}_L(P)$ and $\mathcal{B}_C(P)$ be the families of line blocks and circle blocks of P.

Definition 2.1 (Incidence isomorphisms and inversive equivalence)**.** Let P and P' be finite labelled point sets of the same size.

(i) A *line–circle incidence isomorphism* is a bijection $\sigma : P \to P'$ such that

$$\sigma(\mathcal{B}_L(P)) = \mathcal{B}_L(P'), \qquad \sigma(\mathcal{B}_C(P)) = \mathcal{B}_C(P').$$

Thus membership, maximality, block size, and the distinction between lines and circles are all preserved. The corresponding isomorphism class is called the *line–circle incidence type* of P.

(ii) The *generalized-circle incidence structure* of P is the set system $\mathcal{B}_L(P) \cup \mathcal{B}_C(P)$ after the information specifying whether each block is a line or a circle has been forgotten. An isomorphism of these set systems is a *generalized-circle incidence isomorphism*, and the corresponding isomorphism class is the *generalized-circle incidence type*.

(iii) The sets are *Möbius equivalent* if, after a relabelling, a map

$$z \longmapsto \frac{\alpha z + \beta}{\gamma z + \delta}, \qquad \alpha\delta - \beta\gamma \neq 0,$$

sends P to P' and no point of P to ∞. They are *extended-Möbius equivalent* if one also allows an anti-Möbius map, obtained by replacing z with $\overline{z}$ in the displayed formula.

Möbius and anti-Möbius maps preserve generalized circles, so either geometric equivalence induces a generalized-circle incidence isomorphism. The converse need not hold. Moreover, $c(P)$ itself is not Möbius invariant: a generalized circle through the pole of a transformation becomes a Euclidean line and is then not counted by $c(P)$. This is why both the line–circle incidence structure and the structure obtained after forgetting the line/circle type are needed below.

2.1 Counting from a largest collinear or concyclic subset

Lemma 2.2. *If a largest line contains K points of P, then*

$$c(P) \geq D_L(n,r) := r\binom{K}{2} - \binom{r}{2}\left\lfloor \frac{K}{2} \right\rfloor. \tag{2}$$

If a largest circle contains K points of P, then

$$c(P) \geq D_C(n,r) := 1 + r\left(\binom{K}{2} - \left\lfloor \frac{K}{2} \right\rfloor\right) - \binom{r}{2}\left\lfloor \frac{K}{2} \right\rfloor. \tag{3}$$

Proof. Fix a point q outside the chosen largest line or circle. In the line case, q and any pair on the line determine a circle, and these circles are distinct for fixed q. In the circle case, at most $\lfloor K/2 \rfloor$ pairs on the circle are collinear with q; every other pair gives a new circle, in addition to the largest circle itself. Fix two exterior points q, q'. Every circle counted for both points contains q and q' and meets the chosen largest line or largest circle in a two-point set. The two-point sets belonging to two distinct repeated circles are disjoint: otherwise the circles would share q, q' and a third point and hence coincide. Thus at most $\lfloor K/2 \rfloor$ circles are common to the two lists. The first Bonferroni inequality (sum of list sizes minus the sum of pairwise intersections) now gives both displayed lower bounds; a circle occurring in three or more lists does not invalidate this lower estimate. □

2.2 Inversion and multiplicity identities

Fix $x \in P$ and invert $P \setminus \{x\}$ about x, obtaining a noncollinear configuration Q_x of $q = n-1$ points. Indeed, if Q_x were collinear, its inverse would be either a line through x or a circle through x, so all of P would be collinear or concyclic. Let $t_i(x)$ be the number of connecting lines of Q_x that contain exactly i points. Let $b_i(x)$ be the number of these lines that pass through the original geometric point x, and set

$$u_i(x) = t_i(x) - b_i(x).$$

A connecting line through the inversion centre corresponds to a line in the original configuration. An i-point connecting line avoiding the centre corresponds to a circle through x containing exactly $i+1$ original points. This gives the following identities.

Lemma 2.3. *For every $x \in P$,*

$$\sum_{i=2}^{K-1} \binom{i}{2} t_i(x) = \binom{n-1}{2} \tag{4}$$

and

$$\sum_{i=2}^{K-1} i b_i(x) \leq n-1 \tag{5}$$

For every integer $s \geq 2$, one also has

$$\sum_{i \geq s} b_i(x) \leq \left\lfloor \frac{n-1}{s} \right\rfloor. \tag{6}$$

If ℓ_k and c_k are respectively the numbers of k-point lines and k-point circles in P, then

$$c(P) = \sum_{x\in P} \sum_{i=2}^{K-1} \frac{u_i(x)}{i+1} \tag{7}$$

Moreover,

$$\sum_x b_i(x) = (i+1)\ell_{i+1}, \qquad \sum_x u_i(x) = (i+1)c_{i+1} \tag{8}$$

and

$$\binom{n}{3} = \sum_{k\ge 3} \binom{k}{3} (\ell_k + c_k) \tag{9}$$

In particular, $\sum_x t_i(x)$ is divisible by $i+1$.

Proof. Equation (4) partitions the pairs of Q_x by their connecting line. Distinct connecting lines through x contain disjoint subsets of Q_x, which gives (5). Each line counted on the left of (6) uses at least s points of these disjoint subsets; the number of such lines is an integer, which proves the rounded inequality. A k-point circle becomes, under inversion at each one of its points, a $(k-1)$-point line avoiding the centre. It is therefore counted exactly k times in (7); the same argument for lines gives (8). Finally, every noncollinear triple lies on a unique circle, whereas every collinear triple lies on a unique line, which proves (9). □

2.3 Line-arrangement inequalities and intersection constraints

Elliott's equation (4) is

$$t_2(x) \ge 3 + \sum_{i\ge 4} (i-3)t_i(x), \tag{10}$$

which follows from the Euler relation for a real projective line arrangement [6, Eq. (4)]. When the three highest intersection multiplicities vanish, Hirzebruch's equation (9) gives

$$4t_2(x) + 3t_3(x) \ge 4(n-1) + 4\sum_{i\ge 5} (2i-9)t_i(x). \tag{11}$$

Its original hypotheses are $t_q = t_{q-1} = t_{q-2} = 0$; whenever we use the inequality, these hypotheses are verified from the current upper bound on K [7, §6, Eq. (9)]. Under the corresponding multiplicity hypotheses, the Bojanowski inequality is

$$4t_2(x) + 3t_3(x) \ge 4q + \sum_{i\ge 5} i(i-4)t_i(x), \qquad q = n-1, \tag{12}$$

provided no connecting line contains at least $2q/3$ image points, as stated in the dual form in Pokora's Theorem 2.1 [10, Theorem 2.1].

Lemma 2.4 (Rounded lower bound for the number of connecting lines)**.** *Whenever* (10) *and* (12) *apply,*

$$\sum_{i\ge 2} t_i(x) \ge \left\lceil \frac{q^2+3q+9}{9} \right\rceil. \tag{13}$$

Proof. Add 2/9 of the pair identity (4), 1/3 of (10), and 1/9 of (12). The coefficient of t_i is

$$\frac{2}{9}\binom{i}{2} - \frac{i-3}{3} - \frac{i(i-4)}{9} = 1 \quad (i \ge 5),$$

and direct substitution gives the same coefficient for $i = 2, 3, 4$. The right side is $(q^2 + 3q + 9)/9$. The left side is integral, which gives the ceiling. □

Apply Zhang's Theorem 4.1 to the noncollinear set Q_x and to the external point x (the centre is not an image point) [12, Theorem 4.1]. It gives

$$u_2(x) \geq \left\lceil \frac{n-1}{6} \right\rceil, \tag{14}$$

and the theorem of Csima and Sawyer, for $n > 7$, gives $t_2(x) \geq \lceil 6(n-1)/13 \rceil$ [4, the theorem on pp. 187–188].

Order the connecting lines of Q_x containing at least four points as $L_1, L_2, \ldots$, with nonincreasing sizes. Every initial segment satisfies

$$\sum_{a=1}^{j} |L_a| - \binom{j}{2} \leq n-1, \tag{15}$$

because two distinct lines share at most one point. If a fixed connecting line contains a points, the $a(n-1-a)$ pairs between this line and its complement must belong to other connecting lines. If two fixed lines contain a and b points, then after deleting their possible intersection there remain $(a-1)(b-1)$ cross pairs. Hence

$$\sum_i (i-1)t_i - (a-1) \geq a(n-1-a) \tag{16}$$

and

$$\sum_i t_i - 2 \geq (a-1)(b-1) \tag{17}$$

These are necessary rather than sufficient conditions for geometric realizability, but they eliminate many impossible integer multiplicity vectors before coordinates enter the problem.

We shall also need a constraint involving several inversion centres at once. Let $\mathcal{C}_k$ be a specified family of k-point circles, let $g_k = |\mathcal{C}_k|$, and let $d_k(x)$ be the number of circles in $\mathcal{C}_k$ through x. Since two distinct circles meet in at most two points,

$$\sum_x \binom{d_k(x)}{2} \leq 2\binom{g_k}{2} \tag{18}$$

and, for $k \neq m$,

$$\sum_x d_k(x)d_m(x) \leq 2g_k g_m \tag{19}$$

Lemma 2.5 (A second-moment estimate for circle incidences)**.** *Let $\mathcal{C}_k$ and $\mathcal{C}_m$ consist of g_k k-point circles and g_m m-point circles, respectively. Suppose that a local count Φ_x satisfies*

$$\Phi_x \geq \alpha + \beta d_k(x) + \beta' d_m(x) - \gamma\binom{d_k(x)}{2} - \gamma'\binom{d_m(x)}{2} - \eta d_k(x)d_m(x),$$

where $\gamma, \gamma', \eta \geq 0$. Then

$$\sum_{x\in P} \Phi_x \geq n\alpha + \beta k g_k + \beta' m g_m - 2\gamma\binom{g_k}{2} - 2\gamma'\binom{g_m}{2} - 2\eta g_k g_m. \tag{20}$$

Proof. Double counting gives $\sum_x d_k(x) = kg_k$ and $\sum_x d_m(x) = mg_m$. Sum the local inequality over x and apply (18)–(19) to the terms with nonpositive coefficients. □

Thus one need not enumerate the full distribution of incidence degrees over all centres. It suffices to find a quadratic lower bound of the displayed form for the finite table of local minima. The case of one circle family is obtained by setting $d_m = 0$.

3 A mathematical proof for $n \geq 17$

Put $q = n - 1$, and recall that K is the largest number of original points on a line or a circle. Every connecting line in an inverted configuration Q_x therefore contains at most $K - 1$ image points. Write

$$\Phi_x = \sum_{i=2}^{K-1} \frac{t_i(x) - b_i(x)}{i+1};$$

by (7), $c(P) = \sum_x \Phi_x$.

Lemma 3.1. *Suppose that*

$$3(K-1) < 2(n-1). \tag{21}$$

Then, for every $x \in P$,

$$\Phi_x \geq \frac{K+1}{18K}\binom{q}{2} + \frac{K+1}{2K} + \frac{K-2}{9K}q - \frac{1}{3}\left\lfloor \frac{q}{2} \right\rfloor. \tag{22}$$

Proof. Condition (21) is precisely the maximum-multiplicity hypothesis needed for Bojanowski's inequality

$$4t_2 + 3t_3 \geq 4q + \sum_{i \geq 5} i(i-4)t_i$$

[10, Theorem 2.1]. Multiply the point-pair identity, Melchior's inequality, and this inequality by

$$\alpha = \frac{K+1}{18K}, \qquad \beta = \frac{K+1}{6K}, \qquad \gamma = \frac{K-2}{36K},$$

respectively, and add them. The resulting coefficient of t_i is at most $1/(i+1)$. For $i = 2, 3$ equality holds; the differences between $1/(i+1)$ and the resulting coefficient are

$$\frac{K-5}{30K} \quad (i=4), \qquad \frac{(i-3)(i-2)(K-i-1)}{12K(i+1)} \quad (5 \leq i \leq K-1).$$

They are nonnegative. (When $K = 3, 4$, the $i = 4$ term is absent.) Hence

$$\sum_i \frac{t_i(x)}{i+1} \geq \alpha\binom{q}{2} + 3\beta + 4q\gamma.$$

The connecting lines counted by the $b_i(x)$ pass through the inversion centre and contain pairwise disjoint subsets of the image points. Consequently

$$\sum_i b_i(x) \leq \left\lfloor \frac{q}{2} \right\rfloor, \qquad \sum_i \frac{b_i(x)}{i+1} \leq \frac{1}{3}\left\lfloor \frac{q}{2} \right\rfloor,$$

which proves (22). □

Theorem 3.2. *If $n \geq 17$, every admissible n-point set P satisfies $c(P) \geq F(n)$.*

Proof. First assume (21). Summing (22) over the n inversion centres and subtracting $F(n) - 1$ gives, when n is even,

$$\frac{Kn^3 - 23Kn^2 + 100Kn - 72K + n^3 - 11n^2 + 28n}{36K}, \tag{23}$$

and, when n is odd,

$$\frac{Kn^3 - 23Kn^2 + 94Kn - 54K + n^3 - 11n^2 + 28n}{36K}. \tag{24}$$

Both expressions decrease with K, since their derivative is

$$-\frac{n(n-7)(n-4)}{36K^2}.$$

Condition (21) gives $K \le (2n+1)/3$. At this continuous right endpoint, (23) and (24) become, respectively,

$$\frac{n^4 - 21n^3 + 72n^2 + 20n - 36}{18(2n+1)}, \qquad \frac{n^4 - 21n^3 + 66n^2 + 35n - 27}{18(2n+1)}.$$

For even $n = 18 + 2u$ and odd $n = 19 + 2u$, the two numerators are

$$16u^4 + 408u^3 + 3528u^2 + 11056u + 6156,$$
$$16u^4 + 440u^3 + 4140u^2 + 14472u + 10746,$$

so they are positive for $u \ge 0$. At $n = 17$, integrality gives $K \le 11$, and (24) has minimum $10/33$. Thus $c(P) > F(n) - 1$, and integrality yields $c(P) \ge F(n)$.

It remains to consider $3(K-1) \ge 2(n-1)$. Since $n \ge 17$, this implies $3K \ge n + 12$. Put $r = n - K$. Using $\lfloor K/2 \rfloor \le K/2$, the two estimates in Lemma 2.2 give, respectively,

$$D_L(n,r) \ge \frac{rK(3K-n-1)}{4} = B(n,K) + \frac{rK}{2} - 1 \ge B(n,K)$$

and

$$D_C(n,r) \ge 1 + \frac{rK(3K-n-3)}{4} = B(n,K),$$

where

$$B(n,K) := 1 + \frac{(n-K)K(3K-n-3)}{4}. \tag{25}$$

On the real interval $(n+12)/3 \le K \le n-2$, the derivative of B is a concave quadratic and is positive at the left endpoint, where it equals

$$\frac{2n^2 + 21n - 360}{12} > 0.$$

Because a concave quadratic can cross the horizontal axis from positive to negative at most once on this interval, (B) is either increasing throughout or first increasing and then decreasing. Hence (B) has no interior minimum. Using $F(n) \le (n^2 - 4n + 6)/2$, its margins at the two endpoints are

$$5n - 38, \qquad \frac{(n-7)(n-2)}{2},$$

both positive. For $K = n - 1$, the exact largest-circle expression in (3) equals $F(n)$, while the largest-line expression exceeds it by $\lfloor K/2 \rfloor - 1$. This closes the complementary branch. □

The supplementary algebra audit expands every coefficient difference and both parity polynomials with exact symbolic arithmetic. It does not enumerate n, K, or any line-multiplicity vector.

4 The layer $n = 16$

The local bound (22) remains valid at $n = 16$. For $K = 3, 4, 5, 6$, its local values and the resulting margins above 98 are

K	3	4	5	6
Φ_x	20/3	77/12	94/15	37/6
$16\Phi_x - 98$	26/3	14/3	34/15	2/3.

For (K=9,. . . ,15), the smaller of the applicable line and circle bounds in Lemma 2.2 is, respectively,

$$141, 166, 201, 205, 199, 162, 99.$$

These numbers are all at least (F(16)=99). It remains to treat (K=7,8), which we now do without a finite verification.

Suppose first that $K = 7$, and put $d_x = t_6(x) - b_6(x)$; this is the number of seven-point circles through x. Suppress the centre from the notation and define

$$T = \sum_{i=2}^{6} \binom{i}{2} t_i, \qquad M = t_2 - \sum_{i=4}^{6} (i-3)t_i,$$
$$B = 4t_2 + 3t_3 - \sum_{i=5}^{6} i(i-4)t_i, \quad S = \sum_{i=2}^{6} b_i.$$

The pair identity, Melchior, Bojanowski, and the fact that distinct connecting lines through the inversion centre contain disjoint sets of image points give, respectively,

$$T = 105, \qquad M \geq 3, \qquad B \geq 60, \qquad S \leq 7.$$

A direct comparison of coefficients gives the identity

$$\begin{aligned} \Phi_x + \frac{d_x}{42} &= \frac{7}{108}T + \frac{7}{36}M + \frac{1}{54}B - \frac{1}{3}S + R_7, \\ R_7 &= \frac{t_4}{180} + \frac{b_3}{12} + \frac{2b_4}{15} + \frac{b_5}{6} + \frac{b_6}{6} \geq 0. \end{aligned} \tag{26}$$

Consequently

$$\Phi_x + \frac{d_x}{42} \geq \frac{37}{6}. \tag{27}$$

Let g be the number of seven-point circles. Four such circles cannot pass through one point: after inversion their four six-point lines would have union of size at least $4 \cdot 6 - \binom{4}{2} = 18 > 15$. Thus $d_x \leq 3$, and

$$\sum_x d_x = 7g, \qquad \sum_x \binom{d_x}{2} \leq 2\binom{g}{2} = g(g-1).$$

If $g = 5$ or 6, the pointwise inequality $\binom{d}{2} \geq 2d - 3$ for $0 \leq d \leq 3$ gives the contradictory lower bounds $14g - 48 = 22, 36$, whereas the displayed upper bounds are $20, 30$. The case $g \geq 7$ contradicts $7g = \sum_x d_x \leq 16 \cdot 3$. For $g \leq 3$, summing (27) gives

$$c(P) \geq \frac{296}{3} - \frac{g}{6} = 98 + \frac{4-g}{6} > 98.$$

It remains only to exclude equality when $g = 4$. If no point had $d_x = 1$, then $d_x \in \{0, 2, 3\}$ and $\binom{d_x}{2} \geq d_x/2$; hence the second intersection moment would be at least $14 > 12$. Choose a point with $d_x = 1$. If $c(P) = 98$, equality must hold in every term used in (26). The positive residual and the equality cases of Melchior and Bojanowski then give the unique multiplicity vectors

$$(t_2, t_3, t_4, t_5, t_6) = (14, 12, 0, 4, 1), \qquad (b_2, b_3, b_4, b_5, b_6) = (7, 0, 0, 0, 0).$$

The four five-point lines and one six-point line represented by these vectors have union of size at least $4 \cdot 5 + 6 - \binom{5}{2} = 16$, again impossible on fifteen image points. This proves the case $K = 7$.

Now let $K = 8$. At any point on a chosen largest eight-point line or circle, inversion sends the other seven points of that block to a single seven-point image line. Put

$$O = t_2 - b_2, \qquad I = \sum_{i=2}^{7} (i-1)t_i, \qquad S = \sum_{i=2}^{7} b_i,$$

and define T, M as above with upper index 7. The ordinary-line bound, inequality (5), and incidence counting with the seven-point image line give

$$O \geq 3, \qquad S \leq 7, \qquad I \geq 6 + 7(15-7) = 62.$$

Another coefficient identity is

$$\begin{aligned} \Phi_x &= \frac{1}{80}T + \frac{31}{160}M + \frac{1}{48}O - \frac{5}{16}S + \frac{17}{160}I + R_8, \\ R_8 &= \frac{t_5}{240} + \frac{3t_6}{560} + \frac{b_3}{16} + \frac{9b_4}{80} + \frac{7b_5}{48} + \frac{19b_6}{112} + \frac{3b_7}{16} \geq 0. \end{aligned} \tag{28}$$

It follows that each of the eight centres on the chosen block satisfies $\Phi_x \geq 1017/160$. At every other centre the unconditional bound (22) gives $\Phi_x \geq 145/24$. Therefore

$$8 \cdot \frac{1017}{160} + 8 \cdot \frac{145}{24} - 98 = \frac{71}{60} > 0.$$

Thus $c(16) \geq 99 = F(16)$. The accompanying verifier only expands (26) and (28) and checks the displayed equality and moment calculations with exact rational arithmetic; it contains no optimization model or multiplicity-vector enumeration.

5 The fifteen-point case

We now prove $c(15) = 85$. Let K be the largest number of points of an admissible set P on one line or circle. If $K \geq 8$, Lemma 2.2 gives $c(P) \geq 85$. We may therefore assume $K \leq 7$.

For $x \in P$, invert the other fourteen points about x and dualize the resulting point set. This gives an arrangement of fourteen real projective lines. Write $t_r(x)$ for its number of r-fold vertices, and put

$$\delta_x = t_2(x) - 3 - \sum_{r=4}^{6}(r-3)t_r(x).$$

Melchior's relation identifies δ_x with the total excess of the face degrees above three, so $\delta_x \geq 0$.

The fact that the number of lines is even rules out $\delta_x = 1$. Indeed, the sign of the product of the fourteen homogeneous line equations is well-defined on the projective plane and gives a two-colouring of the faces. If $E = \sum_r r t_r(x)$ is the number of edges and F_+, F_- are the two numbers of faces, then

$$\varepsilon_+ = E - 3F_+, \qquad \varepsilon_- = E - 3F_-$$

are nonnegative, congruent modulo 3, and satisfy $\delta_x = \varepsilon_+ + \varepsilon_-$. Their sum cannot be one.

Define the following integer associated with the centre x:

$$R_x = 136t_2(x) + 93t_3(x) + 60t_4(x) + 30t_5(x) - 2902. \tag{29}$$

We first show that $R_x \geq 0$. The point-pair identity and the definition of δ_x give

$$3t_3 + 7t_4 + 12t_5 + 18t_6 = 88 - \delta_x.$$

Since the maximum vertex multiplicity is six, the hypothesis of Bojanowski's inequality is satisfied, and

$$4t_2 + 3t_3 - 5t_5 - 12t_6 \geq 56.$$

Eliminating t_2, t_3 yields

$$3t_4 + 9t_5 + 18t_6 \leq 44 + 3\delta_x \tag{30}$$

and

$$R_x = 234 + 105\delta_x - 21t_4 - 70t_5 - 150t_6. \tag{31}$$

If $\delta_x \geq 2$, then (30) gives

$$21t_4 + 70t_5 + 150t_6 \leq 25(t_4 + 3t_5 + 6t_6) \leq \frac{25}{3}(44 + 3\delta_x),$$

and hence $R_x \geq (240\delta_x - 398)/3$. Since R_x is an integer, this gives $R_x \geq 28$.

It remains to consider $\delta_x = 0$, when the arrangement is simplicial. Cuntz's complete classification through twenty-seven lines [5, Theorem 1.1 and the fourteen-line entries of the invariant table, p. 696] shows that, after excluding the type with a sevenfold vertex and the near-pencil, the possible vectors $(t_2, t_3, t_4, t_5, t_6)$ are

$$(11, 12, 4, 2, 0), \qquad (9, 16, 4, 1, 0), \qquad (10, 14, 4, 0, 1).$$

Their values of R_x are respectively $10, 80, 0$. Consequently

$$R_x \geq 0, \qquad R_x > 0 \Longrightarrow R_x = 10 \text{ or } R_x \geq 28. \tag{32}$$

Let ℓ_k and c_k denote the numbers of maximal k-point lines and circles, and put $C = \sum_{k=3}^{7} c_k = c(P)$. We use the exact identities

$$\ell_2 + \sum_{k=3}^{7} \binom{k}{2} \ell_k = \binom{15}{2}, \qquad \sum_{k=3}^{7} \binom{k}{3} (\ell_k + c_k) = \binom{15}{3},$$

and

$$\sum_{x \in P} t_{k-1}(x) = k(\ell_k + c_k) \qquad (3 \leq k \leq 7).$$

The Csima–Sawyer theorem gives $\ell_2 \geq \lceil 90/13 \rceil = 7$ [4, the theorem on pp. 187–188]. Direct substitution in the preceding identities gives

$$420C - 35270 = 140(\ell_2 - 7) + 420\ell_4 + 980\ell_5 + 1680\ell_6 + 2520\ell_7 + \sum_{x \in P} R_x. \tag{33}$$

Every term on the right is nonnegative, so $C \geq 84$.

Suppose that $C = 84$. The left side of (33) is then 10. Thus $\ell_2 = 7$, $\ell_4 = \cdots = \ell_7 = 0$, and $\sum_x R_x = 10$. By (32), exactly one centre has local type $(11, 12, 4, 2, 0)$, while the other fourteen have type $(10, 14, 4, 0, 1)$. It follows that

$$\sum_{x \in P} t_2(x) = 11 + 14 \cdot 10 = 151.$$

This is impossible because the local incidence identity for $k = 3$ says that the same sum is $3(\ell_3 + c_3)$. Hence $C \geq 85$. The construction in (1) has $F(15) = 85$, and therefore $c(15) = 85$.

Remark 5.1. The proof does not choose the point subsets of any lines or circles and does not enumerate geometric orbits. The finite input consists only of the three multiplicity vectors in Cuntz's complete fourteen-line table; all remaining steps are integer identities and inequalities.

6 The cases $9 \leq n \leq 14$

This section separates the two larger cases, which are decided at the level of global integer data, from the four smaller cases that require an exact finite verification. No historical search tree or saved orbit list is used.

For $k \geq 3$, let ℓ_k and c_k denote the numbers of maximal k-point lines and circles, and put $m_k = \ell_k + c_k$. Every pair belongs to one maximal line, while every triple belongs either to its maximal line or to its unique circumcircle. Therefore

$$\ell_2 + \sum_{k\ge 3} \binom{k}{2} \ell_k = \binom{n}{2}, \tag{34}$$

$$\sum_{k\ge 3} \binom{k}{3} m_k = \binom{n}{3}, \qquad \sum_{k\ge 3} c_k = c(P). \tag{35}$$

We call $(\ell_3, \dots, \ell_K; c_3, \dots, c_K)$ the *global count vector*; it records only the number of maximal lines and circles of each size, not the subsets of points on them. Lemma 2.2 excludes maximum blocks of size at least $6, 7, 7, 7, 7, 8$ for $n = 9, 10, 11, 12, 13, 14$, respectively. Thus K is at most $5, 6, 6, 6, 6, 7$ in the branches considered below.

The enumeration of these count vectors uses only necessary conditions. Since $K < n-1$, every one-point deletion is admissible, and summing its circle count gives

$$nc(P) - 3c_3 \ge nc(n-1). \tag{36}$$

Inversion followed by Zhang's prescribed-point ordinary-line theorem [12, Theorem 4.1] gives

$$3c_3 \ge n \left\lceil \frac{n-1}{6} \right\rceil. \tag{37}$$

We also use the ordinary-line, Melchior, Hirzebruch–Bojanowski, and Shnurnikov inequalities for the line part of the count vector, and their sums over subsets that are forced to be noncollinear [7, Eq. (9)] [10, Theorem 2.1][11, Theorem 3]. For a subset contained in a maximal line no circle lower bound is imposed; for a subset contained in a maximal circle only that defining circle is imposed. This is the correction needed when deletion averages are used at small order.

We shall also use the following elementary consequence of block intersections. For $N \ge 1$ and $S = aN + b$, $0 \le b < N$, set

$$\Phi_N(S) = N\binom{a}{2} + ab.$$

This is the minimum of $\sum_i \binom{x_i}{2}$ over nonnegative integral x_i with $\sum_i x_i = S$. If a selected family consists of L maximal lines and G maximal circles and I_1, I_2 are its total point and point-pair incidences, then

$$\Phi_n(I_1) \le \binom{L}{2} + 2LG + 2\binom{G}{2}, \tag{38}$$

$$\Phi_{\binom{n}{2}}(I_2) \le LG + \binom{G}{2}. \tag{39}$$

Indeed, two lines meet in at most one point and any other two distinct blocks meet in at most two points. Applying these inequalities to the largest members of each subfamily covers every possible choice. It does not decide which labelled points lie on the selected lines and circles.

6.1 The augmented-line inequality

Fix $x \in P$, invert about x, and dualize the remaining $n-1$ points. Let $t_j(x)$ be the number of multiplicity-j vertices of the resulting arrangement and put

$$\delta_x = t_2(x) - 3 - \sum_{j\ge 4} (j-3) t_j(x).$$

Write $d_k^L(x)$ for the number of maximal k-point lines of P through x.

Lemma 6.1. *For every $x \in P$,*

$$\sum_{k\ge 3} k d_k^L(x) \le n - 1 + \delta_x. \tag{40}$$

Consequently,

$$\sum_{k\ge 3} k^2 \ell_k \le n(n-1) + D, \qquad D = 3m_3 - 3n - \sum_{k\ge 5} k(k-4)m_k. \tag{41}$$

Proof. Let $\mathcal{A}_x$ be the arrangement of $n-1$ dual lines, and adjoin the projective line dual to the inversion centre. A k-point original line through x corresponds to an old vertex of multiplicity $k-1$ on the added line. Suppose the selected old vertices have multiplicities $i_1, \dots, i_s$. The added line has $(n-1) - \sum_j i_j$ new ordinary vertices. Raising the multiplicity of each selected old vertex by one decreases the Melchior defect by one, including the cases of multiplicity two and three. Hence the defect of the enlarged arrangement is

$$\delta_x + (n-1) - \sum_{j=1}^{s} (i_j + 1) = \delta_x + (n-1) - \sum_{k\ge 3} k d_k^L(x).$$

The enlarged arrangement is essential, so Melchior's inequality proves (40). Summing over x, and using $\sum_x d_k^L(x) = k\ell_k$ and the local incidence identities, gives (41). □

6.2 Global exclusion for thirteen and fourteen points

For $n = 14$, the exact integer solutions of the preceding necessary conditions with $c(P) < F(14)$ give 8981 count vectors. Inequality (41) excludes 8980. The remaining vector is

$$(\ell_3, \dots, \ell_7; c_3, \dots, c_7) = (20, 0, 0, 0, 0; 23, 44, 0, 2, 3).$$

Its three seven-point circles have pairwise intersections of size at most two, so their union has at least $3 \cdot 7 - 3 \cdot 2 = 15$ points, a contradiction. Thus $c(14) = F(14)$.

For $n = 13$, the same exact enumeration gives 3962 count vectors below $F(13)$. The augmented-line inequality leaves 137, and (38)–(39) leave 73. We now derive two inequalities that exclude these remaining vectors.

At any point x, the local arrangement has twelve lines and, in the present branch, no vertex of multiplicity greater than five. Write its multiplicity vector as (t_2, t_3, t_4, t_5) and its defect as δ. Then

$$-2\delta + t_4 + 3t_5 \le 6, \qquad -\delta + t_5 \le 1. \tag{42}$$

For the first inequality, the pair identity and Bojanowski's inequality give

$$t_2 + 3t_3 + 6t_4 + 10t_5 = 66,$$
$$4t_2 + 3t_3 \ge 48 + 5t_5,$$

and hence

$$H := 3t_2 - 6t_4 - 15t_5 + 18 \ge 0.$$

Put $E = 2t_2 - 3t_4 - 7t_5$. Since $H + 3\delta = 3(E+3)$, one has $E \ge -3$. Moreover, the pair identity gives $E \equiv 0 \pmod 3$. Equality $E = -3$ would force $H = \delta = 0$. Cuntz's complete list of simplicial twelve-line arrangements, after excluding a six-fold vertex, leaves the types $(8, 10, 3, 1)$ and $(9, 7, 6, 0)$; both have $E = 0$ [5, Theorem 1.1 and the invariant tables]. Therefore $E \ge 0$, which is the first inequality in (42).

For the second inequality, four five-fold vertices would dualize to four five-point lines whose union has at least $20 - \binom{4}{2} = 14$ points; hence $t_5 \le 3$. If $\delta \ge 2$ the claim follows. If $\delta = 0$, Cuntz's two types above have $t_5 \le 1$. The only remaining possible violation is $(\delta, t_5) = (1, 3)$. The pair and defect identities then force $(t_2, t_3, t_4, t_5) = (12, 4, 2, 3)$. But three five-point lines and one four-point line have a union of at least $5+5+5+4-\binom{4}{2}=13$ points, again impossible on twelve points. This proves the second inequality.

Summing (42) over the thirteen centres yields

$$-2D+5m_5+18m_6 \le 78, \qquad -D+6m_6 \le 13. \tag{43}$$

Of the 73 count vectors, 71 violate the first inequality and the other two violate the second. This is an exact substitution into (43); no subsets of labelled points are selected. Hence $c(13)=F(13)$.

6.3 Exact verification for nine through twelve points

For $9 \le n \le 12$, the same global constraints and (41) leave a small but nonempty boundary. We describe the common verification so that every transition can be reproduced. For a marked point x, let $a_k(x)$ and $u_k(x)$ count maximal k-point lines and circles through x. Then

$$t_{k-1}(x)=a_k(x)+u_k(x), \qquad \sum_{j\ge 2}\binom{j}{2}t_j(x)=\binom{n-1}{2}.$$

These local multiplicity vectors obey the published arrangement inequalities used above, the block-union inequalities, and (40). A first integer model records only how many points have each vector and imposes

$$\sum_x a_k(x)=k\ell_k, \qquad \sum_x u_k(x)=kc_k, \qquad \sum_x \delta_x = D.$$

The next two stages still do not assign labelled point subsets. For a block category α, let $d_\alpha(x)$ be its point degree and let $\lambda_\alpha(e)$ be its degree at a point pair e. If k_α is the block size and N_α its multiplicity, then

$$\sum_x d_\alpha(x)=k_\alpha N_\alpha, \qquad \sum_e \lambda_\alpha(e)=\binom{k_\alpha}{2}N_\alpha, \tag{44}$$

$$\sum_\alpha (k_\alpha-2)\lambda_\alpha(e)=n-2. \tag{45}$$

The last identity says that the portions outside e of all maximal blocks through e partition $P\setminus e$. The point and point-pair data are linked by

$$\sum_{y\ne x}\lambda_\alpha(\{x,y\})=(k_\alpha-1)d_\alpha(x).$$

The subsequent intersection count uses, for two block categories, the exact identities

$$N_2=M_2, \qquad N_1=M_1-2M_2, \qquad N_0=Q-M_1+M_2,$$

where M_1 is the sum of intersection sizes, M_2 is the sum of their second binomial coefficients, and N_s counts pairs of distinct blocks meeting in s points. It also counts triples disjoint from a distinguished block. All coefficients are binomial counts, and no floating-point relaxation is used.

Finally, inversion at x determines the complete count vector on $n-1$ image points: blocks through x become image lines with one fewer point, whereas blocks avoiding x become image circles. A local vector is retained only if this child vector survives the next level, after which the local multiplicity data at the n points are balanced again. The recursion stops at $n=9$. There the model selects maximal line and circle subsets of the nine labelled points, requires every triple to lie in exactly one selected block, requires a point pair to lie in at most one selected line, and imposes the already proved lower bounds on every admissible proper subset. Infeasibility of this more permissive abstract model implies geometric infeasibility.

The exact counts in the final replay are:

n	initial	augmented sum	local vectors	point–pair	intersections	block pairs	terminal
9	45	36	23	–	–	–	0
10	180	150	55	26	21	12	1
11	416	160	70	39	28	–	0
12	1354	211	169	55	44	–	0

Here "initial" is the number left from 129, 551, 1215, and 4322 raw count vectors, respectively. The terminal zeroes for $n = 11, 12$ come from the same recursion. It visits, respectively,

$$(28, 72, 222) \quad \text{and} \quad (44, 322, 661, 622)$$

count vectors at orders $(11, 10, 9)$ and $(12, 11, 10, 9)$. Every excluded status is an exact `INFEASIBLE` status; there is no `UNKNOWN`.

For $n = 10$ the recursion leaves the single count vector

$$(\ell_3, \ell_4, \ell_5, \ell_6; c_3, c_4, c_5, c_6; \ell_2) = (10, 0, 0, 0; 10, 20, 2, 0; 15).$$

The four local multiplicity vectors retained by the recursive certificate all have $u_5(x) = 1$ and $a_3(x) \le 3$. Since the two five-point circles have ten point incidences, they therefore partition the ten points into sets A and B. Moreover, the ten three-point lines have thirty point incidences, so $a_3(x) = 3$ at every point; the corresponding retained vector also has $u_4(x) = 8$. Every four-point circle meets each of A and B in two points.

Associate such a circle with its chord in A and its chord in B. The two chords meet on the fixed radical axis of the two five-point circles. No three chords determined by five points on a circle can pass through the same point off that circle: three pairwise disjoint chords would require six endpoints. Thus each set of chords meeting at one radical-axis point has size at most two on either side. The twenty four-point circles attain the resulting maximum, so the chord graph is five disjoint copies of $K_{2,2}$. Up to the action of $S_5 \times S_5$, the six perfect matchings of the Petersen graph give two double orbits. One admits no compatible ten-line design; the other has two complementary designs forming one orbit.

For a representative of that orbit, the ten line incidences give a complete projective parametrization with nonzero parameters a, b, d. Five selected circle-intersection minors are

$$ad + b^2, \quad -ab + ad - b, \quad ab - ad - a - 1,$$
$$-ad + bd - b - d, \quad b(ab + 1).$$

They must all vanish. Adding the second and third relations gives $a + b + 1 = 0$, while the last relation and $b \ne 0$ give $ab + 1 = 0$. Hence $b^2 + b - 1 = 0$. The first relation gives $d = b^2/(b + 1)$; here $b + 1 = -a \ne 0$. Substitution in the fourth relation then gives, after removal of nonzero factors,

$$(b - 1)(2b + 1) = 0,$$

which is relatively prime to $b^2 + b - 1$. The discarded parameter branch makes two labelled points coincide, and every factor divided out in forming the five primitive minors is one of a, d, b or ad, all nonzero. Thus the last $n = 10$ count vector is impossible.

Combining these verifications with the construction in (1) gives

$$c(n) = F(n) \qquad (9 \le n \le 14).$$

Remark 6.2 (Scope of the finite verification)**.** The only labelled search in this section occurs in the nine-point terminal model. Orders ten through twelve use count vectors and recursive consistency of local multiplicities, while orders thirteen and fourteen use only global inequalities. No condition $\sum_j t_j \ge 2r - 4$ is assumed for an arbitrary arrangement. For an arrangement of q lines, Elliott's Theorem 3 is used only when its hypotheses $q \ge 10$ and no $(q - 1)$-fold vertex are explicitly satisfied.

7 The finite cases $4 \le n \le 8$

We shall use the following two elementary facts repeatedly.

Lemma 7.1. *Let A, B, C, D be four points in the real projective plane, no three collinear. Then the three points $AB \cap CD$, $AC \cap BD$, and $AD \cap BC$ are not collinear. Moreover, suppose a circle meets each of two lines in two labelled points. If the lines intersect and directed unit coordinates are measured from their intersection, the products of the two coordinate pairs are equal. If the lines are parallel and a common directed coordinate is used, the sums of the two coordinate pairs are equal. Finally, if three Euclidean circles have three distinct pairwise radical axes, those axes are concurrent in the real projective completion of the plane.*

Proof. A projective change of coordinates sends A, B, C, D to $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$. The three displayed intersections then have coordinates $(1, 1, 0), (1, 0, 1), (0, 1, 1)$; their determinant is -2. For two intersecting lines, use them as oblique coordinate axes. A circle has equation

$$x^2 + y^2 + 2kxy + Dx + Ey + G = 0,$$

so its restrictions to the two axes are monic quadratics with the same constant term G. For parallel lines $y = 0$ and $y = h$, the two restrictions are monic quadratics in x with the same linear coefficient D. Vieta's formulas give the two assertions. For the last assertion, normalize the equations of the three circles to have quadratic part $x^2 + y^2$. The three pairwise differences are linear equations whose sum is zero. Hence the intersection of any two of the corresponding projective lines lies on the third. □

For $n = 4$, either three points are collinear, in which case the remaining point with each of the three pairs on that line gives three distinct circles, or no three points are collinear, in which case the four triples give either one four-point circle or four three-point circles. Admissibility excludes the former alternative, so $c(4) \ge 3$. The construction in (1) attains three. For $n = 5$, four collinear points give six circles through the fifth point, while four concyclic points give at least $1 + \binom{4}{2} - \lfloor 4/2 \rfloor = 5$ circles by Lemma 2.2. In the remaining case every line and circle has at most three points. There are at most two collinear triples: three distinct three-point lines on five points would force two of them to share two points. Hence at least eight noncollinear triples remain; because no four points are concyclic, they determine distinct circles. Thus $c(5) \ge 5$, and (1) attains five.

For six points, the direct estimate of Lemma 2.2 gives at least nine circles whenever four points are collinear or five points are collinear or concyclic. If no four points are concyclic and every line contains at most three points, at most four collinear triples can occur: each point is on at most two three-point lines, so the total number of point–line incidences is at most twelve. Thus at least $20 - 4 = 16$ noncollinear triples determine distinct circles. It remains to consider four points a_0, a_1, a_2, a_3 on a circle Γ, with the other points denoted q, r, under the assumptions that no four are collinear and no five are concyclic. Let τ_q, τ_r count the secants through q, r, let s record whether qr contains an a_i, and let b count circles through q, r and two a_i's. Disjointness of the relevant pairs gives $\tau_q, \tau_r, b \le 2$ and $s \le 1$, while triple counting gives

$$c(P) = 17 - \tau_q - \tau_r - s - 3b.$$

If $c(P) \le 7$, then $b = 2$. Relabel the two disjoint pairs as 01 and 23, and let C_{01} and C_{23} be the circles through q, r, a_0, a_1 and q, r, a_2, a_3, respectively. The three pairwise radical axes of Γ, C_{01}, C_{23} are a_0a_1, a_2a_3, qr; hence these three lines are concurrent by Lemma 7.1, in the projective completion. If $s = 1$, this concurrency makes qr equal to one of the first two lines, producing four collinear points. Thus $s = 0$, and the displayed count with $c(P) \le 7$ forces $\tau_q = \tau_r = 2$. The points q, r must be two diagonal points of the complete quadrilateral on a_0, a_1, a_2, a_3, whereas the same radical-axis relation would make all three diagonal points collinear, contrary to Lemma 7.1. Hence $c(6) \ge 8$, and the construction below attains eight.

For seven points, the largest-block estimate first removes every case with at least five points on a line or circle from a putative configuration with at most ten circles. When all blocks have size at most four, write ℓ_3, ℓ_4 for the numbers of maximal three- and four-point lines, put $\ell = \ell_3 + 4\ell_4$, and let c_4 be the number of four-point circles. Unique ownership of triples gives

$$c(P) = 35 - \ell - 3c_4.$$

There are at most two four-point lines, and the pair identity gives $3\ell_3 + 6\ell_4 \le 21$; hence $\ell \le 11$. Thus $c(P) \le 10$ requires at least five four-point circles. The exact isomorphism classification of such circle families gives two classes with five circles and one class each with six and seven circles. Neither five-circle class has enough compatible collinear triples. The six-circle class has a unique extension with $c(P) \le 10$; its seven three-point lines cover all 21 point pairs, contrary to the Sylvester–Gallai theorem [3, pp. 111–112].

It remains to exclude the seven-circle family. Every compatible extension with $c(P) \le 10$ contains, after relabeling, the lines 016 and 025. Normalize $p_0 = (0, 0)$, $p_1 = (1, 0)$, $p_2 = (0, 1)$, $p_6 = (a, 0)$ and $p_5 = (0, b)$. Since this is an affine normalization, write the common quadratic part of every circle as

$$Q(x, y) = x^2 + 2kxy + hy^2, \qquad h - k^2 > 0.$$

The seven prescribed circle equations first give $a = bh$ and, with $t = (h - a)/(a - 1)$,

$$x_3 = ty_3, \qquad x_4 = -ty_4,$$

and the first two equations then give

$$y_3 = \frac{t + h}{Q(t, 1)}, \qquad y_4 = \frac{a - t}{Q(-t, 1)}.$$

Positive definiteness makes both denominators positive. The last two circle equations reduce to

$$ah - ak - hk + h = 0, \qquad ak - a - h + k = 0.$$

Distinctness gives $a \ne 0, 1$, $b \ne 1$, and $y_3 \ne 0$. Hence the numerator $t + h$ is nonzero; since $t + h = a(h - 1)/(a - 1)$, one has $h \ne 1$. Adding the two displayed equations now gives $a = k$, and the second gives $h = k^2$, contradicting positive definiteness. The exact reductions are checked in Appendix B. Hence $c(7) \ge 11$; the construction below attains eleven.

For eight points, Lemma 2.2 removes every line or circle of size at least five from any putative example with at most sixteen circles. Thus all remaining blocks have size three or four. Let ℓ be the number of collinear triples and let c_4 count four-point circles. Then

$$c(P) = 56 - \ell - 3c_4.$$

In fact the range below twelve can be excluded without a finite search. Write $m = \ell_4 + c_4$ for the number of four-point blocks and let ℓ_3 count three-point lines. The four triples owned by distinct four-point blocks are disjoint, so $m \le 14$, and at most $56 - 4m$ triples remain available for three-point lines. Hence

$$\ell_3 \le \min\{7, 56 - 4m\},$$

where the first bound is the eight-point orchard bound [2, Table I]. Moreover, three four-point lines would have union of size at least $12 - 3 = 9$, since two distinct lines meet in at most one point; thus $\ell_4 \le 2$. Finally, $\ell_2 + 3\ell_3 + 6\ell_4 = 28$ and Melchior's inequality $\ell_2 \ge 3 + \ell_4$ show that $\ell_4 = 2$ implies $\ell_3 \le 3$. Since $\ell = \ell_3 + 4\ell_4$ and $c_4 = m - \ell_4$, the preceding circle-count identity becomes

$$c(P) = 56 - 3m - \ell_3 - \ell_4.$$

If $c(P) \le 11$, then

$$3m + \ell_3 + \ell_4 \ge 45.$$

For $m \leq 11$ the left side is at most $33+7+2=42$. For $m=12$ it is at most 44 when $\ell_4 \leq 1$, and at most 41 when $\ell_4 = 2$. For $m=13$, the bound $\ell_3 \leq 4$, together with the preceding improvement when $\ell_4 = 2$, gives at most 44. For $m=14$ one has $\ell_3 = 0$, again giving at most 44. This proves $c(P) \geq 12$ mathematically.

One uniform finite model now classifies only the five boundary counts from twelve through sixteen. It has two exhaustive branches: a four-point line is fixed up to relabeling, or no four-point line exists. Counts twelve and thirteen each have one incidence class; counts fourteen, fifteen, and sixteen have, respectively, one, three, and three classes. For the twelve-circle class, its two four-point lines and the listed circles give the product or sum relations associated with the three pairings of four distinct coordinates; any two already force a repeated coordinate by Lemma 7.1. In the thirteen-circle class, paired radical axes put all three diagonal points of a complete quadrilateral on the four-point line, contradicting the first assertion of that lemma. For the fourteen-circle class, choose an inversion centre on one selected circle but away from the eight points and its finitely many intersections with the other circles. Exactly that circle becomes a four-point line, and the resulting incidence class is the excluded thirteen-circle class. At fifteen circles, one class is excluded by the exact inversion equations detailed in Appendix B; the other two reduce to the same product/sum relations. At sixteen circles, the two classes containing a four-point line are excluded by the complete-quadrilateral argument. For the remaining class, let E_{ijkl} be the numerator of the Euclidean concyclicity determinant for p_i, p_j, p_k, p_l after the stated coordinate substitution. The exact formulas factor E_{ijkl} into factors already known to be nonzero and one residual polynomial, denoted by F_{ijkl}. Thus the required circle is equivalent to $F_{ijkl} = 0$ in this branch. These residual polynomials satisfy

$$F_{0256} - F_{2357} = 2a(u-b), \qquad F_{0256} + F_{0367} \big|_{u=b} = 2s,$$

where the coordinate normalization has $a \neq 0$ and $s \neq 0$; hence the three required circle equations are inconsistent. These branches exhaust all incidence classes through sixteen circles, so $c(8) \geq 17$. The rational construction below attains seventeen.

Remark 7.2 (What the finite search proves)**.** The integer models classify abstract maximal line and circle blocks subject to unique triple ownership and the intersection bounds for two lines, a line and a circle, or two circles. Feasibility of such a model is not treated as geometric realizability. Every surviving class is therefore followed by a separate exact geometric exclusion. Conversely, an infeasibility conclusion is used only when the solver reports the terminal status `INFEASIBLE`. The statuses `UNKNOWN`, `MODEL_INVALID`, a time limit, or any other nonterminal outcome retain the candidate and make the certificate incomplete; they never exclude a case.

7.1 $c(6) = 8 < 9 = F(6)$

Let $s = \sqrt{3}$ and take

$$p_0 = (7,0), \quad p_1 = (1,0), \quad p_2 = (4,2s),$$
$$p_3 = (1, s/2), \quad p_4 = (1,4s), \quad p_5 = (1/7, 4s/7).$$

The circles containing at least three of the points are exactly

$$0123,\ 0145,\ 025,\ 034,\ 124,\ 125,\ 135,\ 2345. \tag{46}$$

There are $\binom{6}{3} - 3 = 17$ noncollinear triples, and the three four-point circles and five three-point circles in (46) contain $3\binom{4}{3}+5 = 17$ triples. Thus the set determines exactly eight circles. The program `computation/n4_to_8/n6/verify_construction.py` verifies all collinearity determinants, circle equations, and triple covers exactly in $\mathbb{Q}(\sqrt{3})$; the preceding case analysis gives the matching lower bound $c(6) \geq 8$.

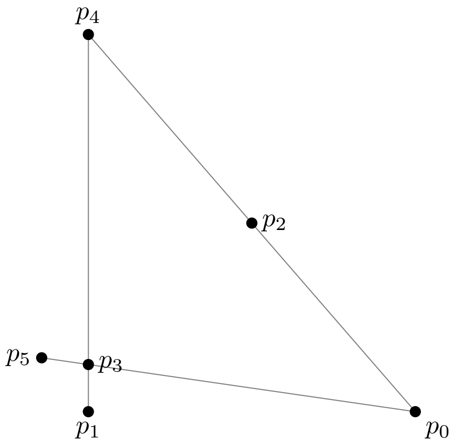

Figure 2: The six-point exceptional construction. Gray segments show its collinear triples; circles are omitted to keep the point diagram legible.

7.2 $c(7) = 11 < 13 = F(7)$

Let $s = \sqrt{3}$ and take

$$\begin{aligned} &p_0 = (0,0), && p_1 = (1,0), && p_2 = (1/2, -s/2), \\ &p_3 = (1/2, s/2), && p_4 = (1/2, -s/6), && p_5 = (3/2, -s/2), \quad p_6 = (-1/2, -s/2). \end{aligned}$$

The circles are exactly

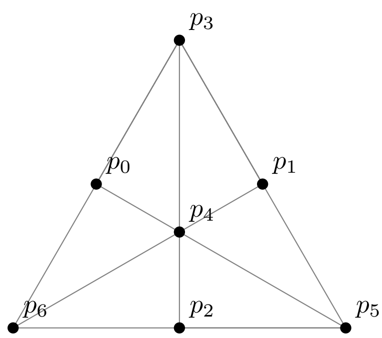

Figure 3: The seven-point construction with eleven circles. Gray segments show all collinear triples; circles are omitted.

$$0134,\ 0156,\ 0235,\ 0246,\ 1236,\ 1245, 012,\ 345,\ 346,\ 356,\ 456.$$

The first six contain four points, and the last five are ordinary circles. Thus the configuration determines exactly eleven circles. The program `computation/n4_to_8/n7/verify_extremal_configuration.py` checks these incidences twice: first over the positive-definite form x^2+xy+y^2, and then after the explicit Euclidean change of coordinates $(x,y) \mapsto (x+y/2, \sqrt{3}\,y/2)$.

It is natural to ask for an intermediate construction with $c(P) = 12 = F(7) - 1$. No such configuration exists. If five or more points lie on one line or circle, Lemma 2.2 gives at least thirteen circles. Otherwise, if c_4 is the number of four-point circles and $\ell = \ell_3 + 4\ell_4$, then $c(P) = 35 - \ell - 3c_4$. The pair bound gives $\ell \le 11$. Equality with twelve circles leaves the cases $(c_4, \ell) = (4, 11), (5, 8), (6, 5), (7, 2)$. The first would leave no ordinary line, contrary to the Sylvester–Gallai theorem [3, pp. 111–112]. The two five-circle packings admit no compatible line family. The five six-circle incidence orbits are excluded by the two-line relations of Lemma 7.1 and four exact coordinate reductions; in one reduction the nominally missing sixth line is forced, producing the eleven-circle configuration above. The unique seven-circle orbit is excluded by the positive-definite calculation already used in the lower bound. These finite assertions are reproduced by `computation/n4_to_8/n7/classify_line_extensions.py`, `verify_circle_count_spectrum.py`, and `verify_q7_exclusion.py`.

7.3 $c(8) = 17 < 19 = F(8)$

We first make the historical Segre example explicit. Take the vertices of two concentric, similarly oriented squares,

$$(\pm 1, \pm 1), \qquad (\pm r, \pm r), \qquad r > 0, \quad r \neq 1.$$

This is the incidence structure obtained by a central projection of a cube along an axis through the centres of two opposite faces, as recorded by Elliott [6, p. 182, discussion following Theorem 2]. It has two four-point lines, ten four-point circles, and eight three-point circles. The number of noncollinear triples is

$$\binom{8}{3} - 2\binom{4}{3} = 48 = 10\binom{4}{3} + 8,$$

so it determines 18 circles.

The following rational construction improves this to 17. Take

$$\begin{aligned}
p_0 &= (0,0), \\
p_1 &= \left(\frac{263}{626}, \frac{2178}{4069}\right), & p_2 &= \left(\frac{263}{313}, \frac{4356}{4069}\right), \\
p_3 &= \left(\frac{789}{626}, \frac{6534}{4069}\right), & p_4 &= \left(\frac{53519}{195938}, \frac{1842342}{1273597}\right), \\
p_5 &= \left(\frac{184032}{458545}, \frac{7245468}{5961085}\right), & p_6 &= \left(\frac{160557}{917090}, \frac{5527026}{5961085}\right), \\
p_7 &= \left(-\frac{25}{313}, \frac{312}{313}\right).
\end{aligned}$$

Its 17 circles are

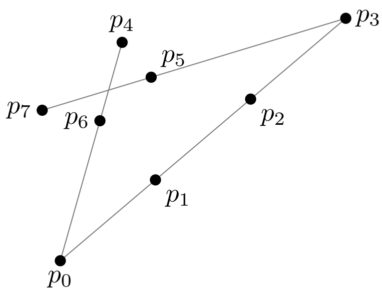

Figure 4: The rational eight-point construction with 17 circles. Gray segments show all maximal collinear subsets; the circles are listed in the text.

$$0145, 0167, 024, 0257, 026, 0347, 0356, 1247, 1256,$$
$$1346, 135, 137, 157, 2345, 2367, 246, 4567.$$

The program `computation/n4_to_8/n8/verify_construction.py` uses rational determinants to verify these circles and checks that all $\binom{8}{3} = 56$ triples belong to exactly one maximal collinear subset or one listed circle. The isomorphism classification of all eight-point incidence structures with at most 16 circles finds none that is realizable in the Euclidean plane. Therefore $c(8) \geq 17$, and hence $c(8) = 17$.

7.4 Incidence types and Möbius moduli of the exceptional configurations

The incidence type of an exceptional configuration should not be confused with its Möbius orbit. The former is finite combinatorial data; the latter retains continuous cross-ratio parameters. The following statement records the complete incidence classification obtained here.

Proposition 7.3. *Up to relabelling, there is exactly one realizable line–circle incidence type for each of $(n, c(P)) = (6, 8), (7, 11), (8, 17)$. At $(n, c(P)) = (8, 18)$ there are sixteen abstract line–circle incidence types, exactly three of which are realizable over $\mathbb{R}$. The three realizable types have, respectively, two four-point lines, one four-point and one three-point line, and two three-point lines. After the line/circle distinction is forgotten, they form one generalized-circle incidence type. Each of the four realizable layers just listed contains continuum many extended-Möbius orbits.*

Proof. The six- and seven-point classifications are the terminal isomorphism reductions used in their lower bounds. At seventeen circles, the eight-point enumeration has seven abstract line–circle incidence types; the exact real-realizability reductions leave only the type whose maximal lines are $0123, 046, 357$. At eighteen circles, triple ownership gives

$$\sum_{L} \binom{|L|}{3} + \sum_{C:\,|C|\geq 4} \left(\binom{|C|}{3} - 1 \right) = 38.$$

Exhaustive backtracking under this identity gives sixteen line–circle incidence types. Twelve of them have one common generalized-circle incidence type that, after a point is sent to infinity, would realize the Fano plane over $\mathbb{R}$. A second generalized-circle incidence type forces the Möbius–Kantor incidence equations; their last determinant is $-(t^2-t+1)$ and has discriminant -3. The remaining generalized-circle incidence type admits exactly the three assignments of its blocks as lines or circles stated above. The seven seventeen-circle records are checked by `computation/n4_to_8/orbit_classification/c17_exclusions/verify_target_records.py`; their exact realizability certificates are in the same directory. The eighteen-circle enumeration and exclusions are reproduced by `computation/n4_to_8/orbit_classification/verify_incidence_types.py`. The positive-dimensional families below, together with the finiteness of the relabeling group, prove the last assertion. □

For completeness, representatives of the three realizable line–circle incidence types at $c(P) = 18$ are as follows. Each list contains all maximal lines and all circle blocks; the index-string convention is that of §2, and C_k denotes the list of k-point circle blocks.

- Type $(4, 4)$ has lines $0145, 2367$ and

$$C_3 : 035, 036, 056, 124, 127, 147, 247, 356,$$
$$C_4 : 0123, 0167, 0246, 0257, 0347, 1256, 1346, 1357, 2345, 4567.$$

- Type $(4, 3)$ has lines $0145, 036$ and

$$C_3 : 035, 056, 124, 127, 147, 247, 356,$$
$$C_4 : 0123, 0167, 0246, 0257, 0347, 1256, 1346, 1357, 2345, 2367, 4567.$$

- Type $(3, 3)$ has lines $035, 124$ and

$$C_3 : 036, 056, 127, 147, 247, 356,$$
$$C_4 : 0123, 0145, 0167, 0246, 0257, 0347, 1256, 1346, 1357, 2345, 2367, 4567.$$

We next make the Möbius quotients explicit. Identify $\mathbb{R}^2$ with $\mathbb{C}$ and use the ordered cross ratio

$$[z_0, z_1; z_2, z_3] = \frac{(z_0 - z_2)(z_1 - z_3)}{(z_0 - z_3)(z_1 - z_2)}.$$

Six points and eight circles. After choosing the displayed incidence labels and applying a similarity, every realization has a unique representative with some (a, v) in

$$\mathcal{U}_6 = \{(a, v) \in \mathbb{R}^2 : a \neq 0, 1,\ v \neq 0,\ a \neq 1 + v^2\}$$

through

$$p_0 = (0,0), \quad p_2 = (1,0), \quad p_4 = (a,0), \quad p_3 = (1,v),$$
$$s = \frac{a}{1+v^2}, \quad t = \frac{a(a-1)}{(a-1)^2+v^2}, \quad p_5 = (s, sv), \quad p_1 = (a + t(1-a), tv).$$

The maximal lines are $024, 035, 134$, and the four-point circles are $0123, 0145, 2345$. Put

$$x = [p_0, p_1; p_2, p_3] = \frac{a-1-v^2}{(a-1)(1+v^2)}, \qquad y = [p_0, p_1; p_4, p_5] = \frac{a-1-v^2}{a-1}.$$

Then

$$v^2 = \frac{y-x}{x}, \qquad a = \frac{y(1-x)}{x(1-y)},$$

and the unlabelled Möbius orbit space is

$$\mathcal{M}_6 = \mathcal{D}_6/\langle R, S\rangle, \qquad \mathcal{D}_6 = \left\{(x,y) : x, y \notin \{0,1\},\ \frac{y-x}{x} > 0\right\},$$

where the group has order six and is generated by

$$R(x,y) = \left(\frac{1}{y}, \frac{1}{x}\right), \qquad S(x,y) = \left(x, \frac{xy-2x+1}{y-x}\right).$$

Every member also has an anti-Möbius symmetry inducing $(p_0p_1)(p_2p_3)(p_4p_5)$; consequently allowing anti-Möbius maps does not further merge the unlabelled six-point orbits.

Seven points and eleven circles. After choosing the displayed incidence labels, every realization likewise has a unique similarity-normalized representative with some

$$(u,v) \in \mathcal{U}_7 = \{(u,v) \in \mathbb{R}^2 : u \neq 0, 1,\ v \neq 0,\ u^2 - u + v^2 \neq 0\}.$$

Set

$$p_3 = (0,0), \quad p_4 = (1,0), \quad p_5 = (u,v), \quad p_6 = \left(u, \frac{u(1-u)}{v}\right),$$

and let

$$p_0 = p_3p_6 \cap p_4p_5, \qquad p_1 = p_3p_5 \cap p_4p_6, \qquad p_2 = p_3p_4 \cap p_5p_6.$$

Explicitly,

$$p_0 = \left(\frac{v^2}{(u-1)^2+v^2}, \frac{-v(u-1)}{(u-1)^2+v^2}\right),$$
$$p_1 = \left(\frac{u^2}{u^2+v^2}, \frac{uv}{u^2+v^2}\right), \qquad p_2 = (u, 0).$$

The maximal lines are $036, 045, 135, 146, 234, 256$ and the six four-point circles are those listed above in the construction. All unlabelled holomorphic Möbius identifications are generated on $\mathcal{U}_7$ by

$$A(u,v) = \left(\frac{u}{u^2+v^2}, \frac{-v}{u^2+v^2}\right), \qquad B(u,v) = \left(\frac{v^2}{u^2-u+v^2}, \frac{-v(u-1)}{u^2-u+v^2}\right).$$

The resulting group has order 24. Thus

$$\mathcal{M}_7^+ = \mathcal{U}_7/\langle A, B\rangle, \qquad \mathcal{M}_7^\pm = \mathcal{U}_7/\langle A, B, C\rangle, \quad C(u,v) = (u, -v),$$

where the extended group has order 48 at a generic parameter.

Eight points. The seventeen-circle type has maximal lines $0123, 046, 357$. It contains the following explicit rational one-parameter family. Put

$$m = \frac{3}{3-b^2}, \qquad r = \frac{3(1+b^2)}{3-b^2}, \qquad t = \frac{b^2-3}{b^2+9}, \qquad O = \left(\frac{1-b^2}{1+b^2}, \frac{2b}{1+b^2}\right).$$

Before inversion take

$$\begin{array}{llll} P_0 = (0,0), & P_1 = (1,0), & P_2 = (1,b), & P_6 = (r,0), \\ P_3 = (1+t(r-1), b(1-t)), & P_4 = P_0P_3 \cap P_1P_2, & P_5 = (m, mb). & \end{array}$$

If I_O is inversion of squared radius one about O, set $p_i = I_O(P_i)$ for $0 \le i \le 6$ and $p_7 = O$. Outside a finite algebraic set of degenerate parameters this has exactly the incidence list displayed for the seventeen-circle construction. The value $b = 13/12$ gives the coordinates used above; $b = 1$ gives a configuration of the same line–circle incidence type that is neither Möbius nor anti-Möbius equivalent to it.

For the eighteen-circle layer, a two-parameter family is given generically by two concentric, similarly oriented, homothetic rectangles

$$(\pm A, \pm 1), \qquad (\pm rA, \pm r).$$

It realizes the type with two four-point line blocks. Inversions at suitable intersections of its generalized circles realize the other two assignments of the blocks as lines or circles in Proposition 7.3. The exact parameter identities and inequivalence tests are in `computation/n4_to_8/orbit_classification/verify_parametric_families.py` and `computation/n4_to_8/orbit_classification/verify_mobius_moduli.py`.

For reference, an unordered four-point generalized circle has the Möbius and anti-Möbius invariant

$$J(\lambda) = \frac{(1-\lambda+\lambda^2)^3}{\lambda^2(1-\lambda)^2},$$

where λ is any ordered cross ratio; hence the multiset of J-values over all maximal four-point generalized circles is an unlabelled invariant. The cited verification first compares this multiset and then checks every remaining incidence relabelling by exact normalized cross ratios. The formulae above give complete Möbius moduli for $n = 6, 7$. For $n = 8$ they prove the existence of continuum many orbits, but we do not claim that the displayed one- and two-parameter families exhaust the full real realization spaces.

8 The variant with no three collinear points

We now prove Corollary 1.3.

Proof. For the general upper bound, take $n-1$ points on a circle and an exterior point that avoids the finitely many secants through pairs of those points. There are no three collinear points, and the construction determines the large circle and one distinct circle for every pair on it. Thus $c_{\mathrm{nc}}(n) \le 1 + \binom{n-1}{2}$.

We first prove the matching lower bound for $n \ge 11$. Here K is the maximum number of concyclic points, because every line contains at most two points. If $3(K-1) < 2(n-1)$, the proof of Lemma 3.1 applies with $b_i(x) = 0$ and gives

$$\Phi_x \ge \frac{K+1}{18K}\binom{n-1}{2} + \frac{K+1}{2K} + \frac{(K-2)(n-1)}{9K}.$$

After summing over x and subtracting $1 + \binom{n-1}{2}$, the margin is

$$\frac{(n-4)\{Kn^2 - 13Kn + 18K + n^2 - 7n\}}{36K}.$$

It decreases with K. At $K = (2n+1)/3$ it equals

$$\frac{(n-4)(n-1)(n^2-10n-9)}{18(2n+1)},$$

which is positive for $n \geq 11$.

Suppose instead that $3(K-1) \geq 2(n-1)$. An exterior point is not collinear with a pair on a largest circle. The proof of Lemma 2.2 therefore improves to

$$c(P) \geq 1 + (n-K)\binom{K}{2} - \binom{n-K}{2}\left\lfloor \frac{K}{2} \right\rfloor \geq 1 + \frac{(n-K)K(3K-n-1)}{4}.$$

On $(2n+1)/3 \leq K \leq n-1$, the derivative of the last expression is a concave quadratic, positive at the left endpoint and negative at the right endpoint. Hence its minimum is attained at an endpoint. Its margins above $1 + \binom{n-1}{2}$ there are, respectively,

$$\frac{(n-4)(n-1)(2n-9)}{36} \quad \text{and} \quad 0.$$

This proves the result for $n \geq 11$.

The remaining orders require only short arguments. For $n = 4$, the four triples determine four distinct circles, since putting two triples on one circle would make all four points concyclic. For $n = 5$, at most one four-point circle can occur, so partitioning the ten triples among the circles gives at least $10 - 3 = 7$ circles. For $n = 6$, if there is a five-point circle, the improved largest-circle bound gives eleven. Otherwise every nonordinary circle has four points. Two such circles share at most two points, so their complementary pairs in the six-point set are disjoint; hence there are at most three four-point circles. Triple counting again gives $c(P) \geq 20 - 3 \cdot 3 = 11$. For $n = 7$, the only potentially smaller case would consist of seven four-point circles. Up to relabelling these are the complements of the lines of the Fano plane. Inverting at one point gives the four lines $123, 145, 246, 356$ and the three circles $1256, 1346, 2345$. Normalize

$$p_1 = (0,0), \quad p_2 = (1,0), \quad p_3 = (t,0), \quad p_4 = (u,v), \quad p_5 = a(u,v),$$

with $p_6 = p_2p_4 \cap p_3p_5$. After nonzero incidence factors are removed, the three circle determinants are

$$-a(u^2+v^2) + 2tu - t, \qquad a(u^2+v^2) - 2au + t, \qquad -a(u^2+v^2) + t.$$

They imply successively $u = 1$, $v = 0$, and $a = t$, contradicting noncollinearity. The finite classification up to relabelling is reproduced by `computation/no_three_collinear/verify_no_three_collinear.py`; hence $c_{\mathrm{nc}}(7) = 16$.

For $n = 9$, if every circle has at most four points and c_4 circles have four points, then $c(P) = \binom{9}{3} - 3c_4$. To bound c_4, fix a point. The four-point circles through it induce three-subsets of the other eight points with no repeated pair, so their number is at most $\lfloor (8/3)\lfloor 7/2 \rfloor \rfloor = 8$. Summing this bound over the nine points and dividing by four gives

$$c_4 \leq \left\lfloor \frac{9}{4} \left\lfloor \frac{8}{3} \left\lfloor \frac{7}{2} \right\rfloor \right\rfloor \right\rfloor = 18,$$

so $c(P) \geq 30$. A circle with at least five points is covered by the improved largest-circle bound, whose minimum is the required 29.

For $n = 10$, the same largest-circle bound deals with $K \geq 6$. If $K \leq 5$, invert at x and write (t_2, t_3, t_4) for the multiplicities of connecting lines among the nine image points. Pair counting, Melchior's inequality, and Hirzebruch's inequality give

$$t_2 + 3t_3 + 6t_4 = 36, \qquad t_2 \geq 3 + t_4, \qquad t_2 \geq 2t_4.$$

Since $t_2 \equiv 0 \pmod 3$, direct substitution for $0 \le t_4 \le 4$ yields

$$\frac{t_2}{3} + \frac{t_3}{4} + \frac{t_4}{5} \ge \frac{18}{5},$$

with equality only at $(t_2, t_3, t_4) = (6, 4, 3)$. If $c(P) \le 36$, equality must hold at all ten inversion centres. Consequently there would be six five-point circles, each point would lie on three of them, and every pair of these circles would meet in exactly two points. The resulting 2-$(6, 3, 2)$ incidence design has one relabelling orbit, as verified by the exact finite enumeration in `computation/no_three_collinear/verify_no_three_collinear.py`.

Invert at a point incident with circles $0, 1, 4$. The remaining nine points lie four at a time on the three sides of a triangle. After an affine normalization, write the two additional points on the three sides as

$$(u, 0), (v, 0), \qquad (0, s), (0, t), \qquad (p, 1-p), (r, 1-r),$$

and retain the general positive-definite quadratic part $x^2 + 2kxy + hy^2$. The first two five-point-circle equations give $hs = uv$ and $hst = u$, hence $t = 1/v$ and $h = uv/s$. Eliminating k from the remaining equations gives

$$p(s+1) = 1, \qquad r(u+1) = u, \qquad (v-1)(su-1) = 0.$$

Distinctness gives $v \ne 1$, so $su = 1$ and therefore $p = r$, again a repeated point. Thus $c_{\mathrm{nc}}(10) = 37$.

It remains to determine the exceptional order eight. Begin with the two concentric squares $(\pm1, \pm1), (\pm2, \pm2)$, which determine eighteen circles and two maximal lines. Invert about $z = (0, 10)$. The point z lies on none of these twenty generalized circles, so all twenty become circles and the image has no collinear triple. Explicitly, the image points are

$$(1/82, 811/82), (1/122, 1209/122), (-1/82, 811/82), (-1/122, 1209/122),$$
$$(1/34, 168/17), (1/74, 367/37), (-1/34, 168/17), (-1/74, 367/37).$$

Figure 5 shows a translated and uniformly rescaled copy of these exact coordinates. With the index-string convention of Section 2, its twenty circles are

$$0123, 0145, 0167, 0246, 0257, 0347, 035, 036, 056, 124,$$
$$1256, 127, 1346, 1357, 147, 2345, 2367, 247, 356, 4567.$$

Hence $c_{\mathrm{nc}}(8) \le 20$.

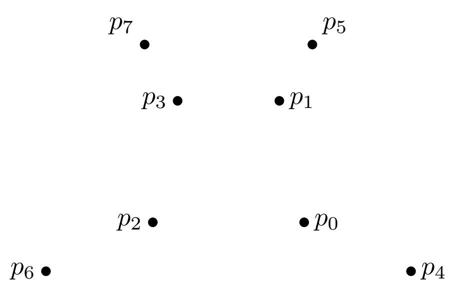

Figure 5: The eight-point construction for the no-three-collinear variant. Only the points are drawn; displaying all twenty circles would obscure the configuration.

For the lower bound, circles of size at least five are again excluded by the largest-circle estimate. If all circles have size at most four, then $c(P) = 56 - 3c_4$. Theorem 1.2 gives $c(P) \ge 17$, so a value below twenty would have to be seventeen. The exact four-subset classification has one line-free class at that value. After inversion and relabelling it has the six lines

$$036, 045, 135, 146, 234, 256$$

and the seven required circles

$$0134, 0156, 0235, 0246, 1236, 1245, 3456.$$

Normalize $p_0 = (0,0)$, $p_1 = (1,0)$, $p_3 = (0,1)$, and write $p_6 = (0,a)$ and $p_5 = (b, 1-b)$. The line incidences determine the other points. For the general quadratic part $x^2 + 2kxy + hy^2$, the first two circle conditions give

$$h = -\frac{b}{a-b+1}, \qquad k = \frac{1-b}{a-b+1}.$$

After these substitutions, the numerator of the 3456 circle determinant is

$$2ab^2(a-1)^2(b-1)(ab-b+1),$$

whose factors are all nonzero by point distinctness and the stated line incidences. The seventeen-circle class is therefore not realizable without a collinear triple. This proves $c_{\mathrm{nc}}(8) = 20$ and completes the proof. □

Remark 8.1. The packing classifications and every polynomial identity in the preceding proof, as well as the rational eight-point construction, are reproduced by `computation/no_three_collinear/verify_no_three_collinear.py`. The script uses exact integer and symbolic arithmetic; no numerical root decision is used.

A The earlier argument for $n \geq 92$

This section records the intermediate argument that first emerged from the Eureka computations. It is logically superseded by the proof for $n \geq 17$ in the main text, but it explains how a numerical threshold was converted into a short mathematical proof before the estimates were strengthened further.

Proposition A.1. *Every admissible n-point set with $n \geq 92$ determines at least $F(n)$ circles.*

Proof. Let K be the size of a largest line or circle block and put $r = n - K$. We divide according to whether K is at least $3n/8$.

Suppose first that $K \geq 3n/8$. The two quantities in Lemma 2.2 satisfy

$$D_L(n,r) - D_C(n,r) = r\left\lfloor \frac{K}{2} \right\rfloor - 1 \geq 0.$$

Using $\lfloor K/2 \rfloor \leq K/2$ in the circle-block expression therefore gives, in either largest-block case,

$$c(P) \geq B(n,K) := 1 + \frac{(n-K)K(3K-n-3)}{4}. \tag{47}$$

For $K = n-1$, the exact value of $D_C(n,1)$ is $F(n)$. We may consequently assume $3n/8 \leq K \leq n-2$. On this interval

$$\frac{\partial^2 B}{\partial K^2} = \frac{-18K + 6n + 6}{4} < 0,$$

so $\partial B/\partial K$ is strictly decreasing. Hence B has no interior minimum, and it is enough to consider the endpoints. Since

$$F(n) \leq 1 + \frac{(n-2)^2}{2}, \tag{48}$$

the two endpoint margins are

$$B\left(n, \frac{3n}{8}\right) - 1 - \frac{(n-2)^2}{2} = \frac{15n^3 - 1384n^2 + 4096n - 4096}{2048}$$

and

$$B(n, n-2) - 1 - \frac{(n-2)^2}{2} = \frac{(n-2)(n-7)}{2}.$$

The first numerator, after writing $n = 92 + u$, becomes

$$15u^3 + 2756u^2 + 130320u + 338880,$$

and the second margin is plainly positive. This proves the required bound in the first case.

Now suppose that $K < 3n/8$. Fix an inversion centre $x \in P$, and put $N = n - 1$ and $M = K - 1$. Every connecting line of the inverted set contains at most M image points. If t_i denotes the number of its i-point connecting lines, set

$$a_M = \begin{cases} 6, & 2 \le M \le 9, \\ \dfrac{\binom{M}{2}}{M-3}, & M \ge 10, \end{cases} \qquad y_M = \frac{1}{a_M + 1}, \qquad z_M = 1 - y_M.$$

The definition of a_M gives

$$y_M \binom{i}{2} \le z_M(i-3) \qquad (4 \le i \le M).$$

Combining the point-pair identity with Melchior's inequality exactly as in Section 2 yields

$$t_2 + t_3 \ge y_M \binom{N}{2} + 3z_M = 3 + y_M \left(\binom{N}{2} - 3 \right). \tag{49}$$

At most $\lfloor N/2 \rfloor$ of these two- or three-point lines pass through the inversion centre. Every remaining line corresponds to a circle containing the centre and altogether three or four points of P; after summing over all centres, each such circle has been counted at most four times. Thus

$$c(P) \ge \frac{n}{4} \left[3 + y_M \left(\binom{n-1}{2} - 3 \right) - \left\lfloor \frac{n-1}{2} \right\rfloor \right]. \tag{50}$$

Because $K < 3n/8$ and K is integral,

$$M = K - 1 \le \frac{3n-9}{8} =: M_0.$$

The function

$$g(s) = \frac{2(s-3)}{(s+3)(s-2)}$$

is decreasing for $s \ge 6$. For $M \ge 10$ one has $y_M = g(M)$, while for $M \le 9$ one has $y_M = 1/7 = g(9)$. Consequently

$$y_M \ge g(M_0) = \frac{16(n-11)}{(n+5)(3n-25)}.$$

Replacing the floor in (50) by $(n-1)/2$ and using (48), the resulting lower bound minus $F(n)$ is at least

$$\frac{n^4 - 105n^3 + 787n^2 - 1931n + 3000}{8(n+5)(3n-25)}.$$

For $n = 98 + u$ the numerator is

$$u^4 + 287u^3 + 27541u^2 + 891829u + 783766,$$

so the expression is positive for every $n \ge 98$.

It remains only to check $92 \le n \le 97$. For each row below, K_0 is the largest integer strictly below $3n/8$, $M_0 = K_0 - 1$, and Δ_n is the right-hand side of (50), evaluated at M_0, minus $F(n)$. Since the bound in (49) decreases with M, this is the least margin in the second case.

n	92	93	94	95	96	97
K_0	34	34	35	35	35	36
M_0	33	33	34	34	34	35
Δ_n	43	88	$\frac{20371}{1184}$	$\frac{9225}{148}$	$\frac{4908}{37}$	$\frac{29823}{836}$

All six margins are positive, completing the proof. □

Remark A.2. No incidence structure is enumerated in this argument. The displayed table is only a check of six exact rational endpoint inequalities. The proof in the main text replaces this early threshold by $n \ge 17$, and the subsequent sections treat $n = 16$ and $n = 15$.

B Exact verification and reproduction

The audited material is reproduced by two independent entry points. From the `Erdos506` directory, run

`computation/verify_certified_results.py`

```
python computation/verify_certified_results.py
```

This command rechecks the mathematical identities for $n \ge 15$, the exact constructions and classifications for $n \le 8$, and Corollary 1.3. The six intervening layers are reproduced separately by

```
cd computation/n9_to_14
python run_verification.py --jobs 8 --seconds 240
```

as described below. Neither entry point accepts random search, floating-point feasibility, a timeout, or an unknown solver status as an exclusion certificate. Both require CPython 3.13, matching the audited binary modules; SymPy performs exact polynomial algebra, and the project-local OR-Tools version 9.15.6755 is pinned and loaded only through `computation/runtime_dependencies.py`.

B.1 The mathematical range

File	Independently checked assertion
`computation/n_ge_17/verify_n_ge_17.py`	expands the coefficient differences in Lemma 3.1, the two parity polynomials, the endpoint margins, and the largest-block algebra, all over the rationals
`computation/n16/verify_n16.py`	expands the two identities in the $K = 7, 8$ branches, checks all residual coefficients, the multiplicities forced at equality, the sums of pairwise circle-intersection counts, and the direct $K = 9, \ldots, 15$ bounds
`computation/n9_to_14/verify_largest_block_reduction.py`	checks the closed-form line and circle bounds for every largest-block size $K \ge 8$ in the fifteen-point case

`computation/n15/verify_n15.py`	checks the local residual values from Cuntz's fourteen-line table, the positive-defect estimate, the coefficients of (33), and the final congruence obstruction; it also records the SHA-256 digest used to identify the cited source PDF

These scripts verify algebra in proofs already given in the text. The fifteen-point check reads only the three published multiplicity vectors listed in the proof; it does not choose labelled point subsets or enumerate orbits or coordinates.

The arithmetic and finite-sum implications used for $n \geq 15$ are also formalized in

`lean/Erdos506Nge15.lean`.

The file is compiled against the pinned Lean 4 and Mathlib toolchain. It contains no `sorry`, `admit`, custom axiom declaration, or unsafe definition. Its scope is deliberately explicit: the Euclidean and real-projective incidence theorems cited in the text are hypotheses of the corresponding Lean statements, while every algebraic consequence drawn from those hypotheses is checked by the kernel. Thus this file is a formalization of the arithmetic-combinatorial layer of the proof, not a formal construction of the underlying projective geometry.

The analogous arithmetic-combinatorial formalization of Corollary 1.3 is

`lean/Erdos506NoThreeCollinear.lean`.

It kernel-checks the six- and nine-point packing deductions, the ten-point local equality case, both large-order endpoint factors, the eight-point congruence gap, and the final case split for every $n \geq 4$. The accompanying `lean/compile_audit_no_three_collinear.json` records a successful warning-as-error build with Lean 4.29.1 and Mathlib commit `5e932f97`; the source contains no proof placeholder or added axiom. As above, Euclidean incidence lemmas and exhaustive finite-classification outputs remain explicit hypotheses rather than silently postulated definitions.

B.2 Nine through fourteen points

The source files for these six layers are in

`computation/n9_to_14/`.

From that directory run

```
python run_verification.py --jobs 8 --seconds 240
```

The number of workers may be changed without changing the models. Every individual solver uses a fixed random seed; a timeout has status `UNKNOWN` and is retained. A successful replay ends with `certificates/manifest.json`, whose seven terminal checks are `true` and whose status is `PASS`. If a replay is interrupted after all terminal certificates have been written, the same terminal and SHA-256 audit can be rerun with `audit_manifest.py`. The active lower-bound table contains $c(7) = 11$. All certificates in the present package were regenerated after that correction; in particular, the twelve-point recursion now has 622, rather than 617, nine-point child count vectors.

The files are grouped by mathematical role as follows. Some filenames retain implementation terminology: in those names, `signature` means the global count vector defined in Section 6, `profile` means the degrees attached to point pairs, and `support` means an actual subset of labelled points. These words are not additional mathematical notions.

File	Role in the verification

`signature_filters.py`	enumerates all global count vectors from (34)–(35), the largest-block bounds, deletion averages, and the cited line-arrangement inequalities
`projected_arrangement_model.py`	generates all numerical local multiplicity types and their line/circle splits; the final source contains no historical line-deletion obstruction table
`verify_local_type_filter.py`	checks exact unlabelled sums of the local multiplicity vectors and writes `nN_01_signature_filter.json`
`verify_augmented_line_melchior.py`	checks first (41), then the sums over all points required by (40); it writes certificates 02 and 03
`filter_by_classified_split_endpoints.py`	checks the consistency identities linking point degrees and point-pair degrees in (44)–(45); the runner explicitly disables both optional finite catalogues
`pair_profile_generator.py`, `verify_pair_moment_filter.py`	generate all point-pair degree data and impose the exact sums of intersection sizes and of their second binomial coefficients, together with the disjoint-triple counts; they write certificate 05
`block_intersection_rows.py`, `block_row_existence_filter.py`	at $n = 10$, impose the three exact triple strata around each represented block and the bound on matchings between two blocks, reducing 21 count vectors to 12
`verify_recursive_inversion_rows.py`	maps each local multiplicity vector to the full count vector one order lower, balances all point totals at each level, and at the nine-point base selects the maximal lines and circles as subsets of the labelled points
`verify_n10_final_radical_axis.py`	reconstructs the two matching orbits, the unique surviving line-design orbit, and the five exact minors that exclude the final ten-point count vector; it first reads certificate 07, records its SHA-256 digest, and checks that the retained rows force the two five-point circles to partition the point set
`verify_n13_n14_global_inequalities.py`	independently regenerates the 3962 and 8981 raw count vectors and checks the global exclusions and (43); it does not select labelled point subsets
`run_verification.py`, `audit_manifest.py`	execute the stages in order and check all terminal statuses, the unique intermediate $n = 10$ count vector, file sizes, and SHA-256 digests

The terminal certificate paths are

$n = 9$
: `certificates/n9_04_exact_support.json`.

$n = 10$
: `certificates/n10_07_recursive.json` and `certificates/n10_08_final_geometry.json`.

$n = 11$
: `certificates/n11_06_recursive.json`.

$n = 12$
: `certificates/n12_06_recursive.json`.

$n = 13, 14$
: `certificates/n13_n14_global_inequalities.json`.

The intermediate filenames have the numerical prefixes displayed in the table and are consumed in that order. Each stage after 01 records the path and SHA-256 digest of its immediate input whenever it reads one. The final manifest records forty-two source and certificate files. Thus a reader may delete the `certificates` directory and regenerate the complete chain from the listed source files.

In the audited four-worker replay, the full chain took 348.52 seconds of wall time. The largest labelled base model had 840 Boolean variables and 546 constraints, and the slowest individual recursive solve took 6.62 seconds. No solve approached the 240-second limit. These figures are reported only to indicate scale; they are not assumptions in any certificate.

The reproduction uses CPython 3.13 because the audited binary modules are its CPython 3.13 builds. The pinned integer solver is OR-Tools 9.15.6755, loaded through `computation/runtime_dependencies.py`. SymPy is used only for the exact ten-point polynomial calculations. Floating-point feasibility, random search, and a solver status other than explicit infeasibility are never used to exclude a case.

B.3 Six and seven points

`computation/n4_to_8/n6/verify_construction.py` checks the six-point coordinates, maximal lines, circle equations, and triple ownership exactly in $\mathbb{Q}(\sqrt{3})$. The finite parameter check `computation/n4_to_8/n6/verify_lower_bound.py` reproduces the last case of the six-point lower-bound argument.

For seven points the reproduction order is:

1. `computation/n4_to_8/n7/classify_quad_packings.py` classifies the families of at least five four-point circles. It returns two classes with five circles and one class each with six and seven circles.
2. `computation/n4_to_8/n7/classify_line_extensions.py` proves that only the six- and seven-circle families can have $c(P) \leq 10$. It checks that the unique six-circle candidate has no ordinary line and that every seven-circle candidate contains the common two-line core used in the proof.
3. `verify_q7_exclusion.py` checks the positive-definite quadratic-form reduction in the text, while `verify_extremal_configuration.py` verifies the eleven-circle construction using exact Euclidean circle equations.
4. `verify_circle_count_spectrum.py` reruns the incidence classification at circle counts eleven and twelve and checks the four symbolic reductions used to verify the eleven-circle layer and prove nonexistence at twelve circles.

The quadratic-form script asserts the two rational factorizations used in the proof and records every nonzero denominator. It performs no root approximation and no search over coordinates. This formulation is invariant under the affine normalization: the former standard-circle normalization, which did not retain the cross term and the second diagonal coefficient, is not used. All paths in this paragraph are under `computation/n4_to_8/n7/`.

B.4 Eight points

All paths below are under `computation/n4_to_8/n8/`.

1. Lemma 2.2, audited by `computation/n9_to_14/verify_largest_block_reduction.py`, first excludes every line or circle containing at least five points. The counting argument in the proof first excludes circle counts below twelve. `classify_boundary_layers.py` then constructs the two exhaustive branches for every remaining circle count $12, \dots, 16$: a four-point line is fixed up to relabeling, or no four-point line exists. Besides triple ownership and the intersection bounds, the model uses only the orchard bound $\ell_3 \leq 7$, the identity $\ell_2 = 28 - 3\ell_3 - 6\ell_4$, and Melchior's inequality $\ell_2 \geq 3 + \ell_4$. Full relabeling-orbit blocking proves completeness. The certificate `boundary_layer_classification.json` records respectively $1, 1, 1, 3, 3$ classes for counts $12, \dots, 16$.
2. The geometric exclusions are independent of the combinatorial solver. `pattern1_inversion_relation.py` and `pattern1_inversion_circles.py` verify the exceptional fifteen-circle inversion equations. The two files `no_four_line_inversion_relations.py` and `no_four_line_key_equations.py` verify the exceptional sixteen-circle equations. The remaining classes are excluded in the text by the radical-axis assertion in Lemma 7.1. For the exceptional fifteen-circle class, inversion at the intersection of its two prescribed lines gives axes on which $0, 1, 2 = (x_0, 0), (x_1, 0), (x_2, 0)$ and $4, 5 = (0, y_4), (0, y_5)$. The Euclidean quadratic form is $x^2 + y^2 + 2kxy$, where $k^2 \neq 1$. The collinearity and first circle equations verified by the two scripts give
$$R = x_0x_1 - 2x_0x_2 + x_1x_2 = 0, \qquad x_0x_2 = y_4y_5.$$
If $x_0x_2 = y_4^2$, then $y_4 = y_5$, so this factor is nonzero. The next two circle equations therefore reduce, with $A = x_0x_2 + y_4^2$ and $B = y_4(x_0 + x_2)$, to $A = kB$ and $kA = B$. Thus $A = B = 0$, whence $x_0 + x_2 = 0$; substituting in $R = 0$ gives $2x_0^2 = 0$, contrary to distinctness. This is the complete geometric consequence of the symbolic output used for that class.
3. Finally, `verify_construction.py` verifies the rational seventeen-circle construction by exact determinants and checks ownership of all 56 triples.

B.5 Exceptional incidence types and Möbius parameters

All paths in this paragraph are under `computation/n4_to_8/orbit_classification/`. The command sequence

```
python verify_incidence_types.py
python verify_parametric_families.py
python verify_mobius_moduli.py
python family_examples.py
```

checks, respectively, the $c(8) = 18$ incidence classification, the symbolic line and circle identities in every displayed family, the complete finite quotient actions for $n = 6, 7$, and exact inequivalence of selected family members. The directory `c17_exclusions/` contains the seven $c(8) = 17$ incidence records, the rational verifier for the realized type, and the symbolic or Gröbner-basis exclusion of each of the other six types. The file `README.md` gives the reproduction order and distinguishes record regeneration from verification of the released records.

B.6 The no-three-collinear variant

The single file `computation/no_three_collinear/verify_no_three_collinear.py` independently checks all finite and algebraic assertions in Corollary 1.3. It counts the twenty circles of the inverted two-square construction, expands both large-n endpoint margins, enumerates the unique seven- and eight-point four-subset packings used in the proof, classifies the 2-$(6, 3, 2)$ design in the ten-point equality branch, and verifies the coordinate eliminations. The enumerations are over finite labelled subsets and are then compared with the full relabelling orbits. All coordinate calculations are exact over the rationals or in the symbolic polynomial ring.

Remark B.1 (Independent replay)**.** The classifiers construct their constraints from the full labelled sets of triples, four-subsets, and five-subsets; the stored JSON files are outputs, not axioms. Orbit blocking is checked by applying all point permutations. Therefore a reader can delete the generated JSON files and reproduce them from the scripts. The dependency loader checks the pinned solver version. The classifiers use one search worker where isomorphism completeness matters. Only the explicit status `INFEASIBLE` excludes a candidate; every other solver status retains it, and any nonterminal status makes the corresponding certificate incomplete.

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